

Due: Thursday March 3rd, latest at 2PM

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1. Assume the electric field has constant magnitude E_0 and point in the x direction, i.e., it is given by $\mathbf{E} = E_0 \hat{i}$.

We wish to calculate the electric potential difference, $V_B - V_A$, between two points on the x - y plane: the point A with coordinates $(0, 3)$ and the point B with coordinates $(4, 0)$. We are ignoring the z -coordinate for this problem. Recall that the potential difference is given by the line integral $V_B - V_A = - \int \mathbf{E} \cdot d\mathbf{l}$, with the integral proceeding along any curve starting at point A and ending at point B .

- (a) [2 pts] Sketch the situation, showing the x and y axis, the two points A and B , and the direction of the electric field.
 - (b) [3 pts] Consider the straight line pointing from A to B . Any element $d\mathbf{l}$ on this line is at the same angle with the direction of the electric field. Calling this angle θ , we obtain $\mathbf{E} \cdot d\mathbf{l} = E_0 |d\mathbf{l}| \cos \theta$. Calculate $\cos \theta$ based on your drawing.
 - (c) [6 pts] Calculate $V_B - V_A = - \int \mathbf{E} \cdot d\mathbf{l}$, integrating along the straight line pointing from A to B . The value of $\cos \theta$, calculated above, might be helpful.
 - (d) [6 pts] Consider now a path consisting of two segments: a line from A to point C at $(4, 3)$, and a line from point C to point B . Draw the situation. Only one of these (AC or CB) segments contribute to the line integral for $V_B - V_A$. Which one? Calculate $V_B - V_A$ by integrating along this segment. You should of course obtain the same answer as what you already obtained by integrating along the straight line AB .
2. Recall that the current through a surface (e.g., through a cross-section of a current-carrying wire) is the surface integral of the current density.
 - (a) [6 pts] A current I is uniformly distributed over a wire of circular cross-section, with radius R . Find the current density J , as a function of I and R .
 - (b) [6 pts] Now imagine that the current density is larger near the outer reigin of the wire cross-section than at the center: $J = kr$, where r is the radial distance from the center of the wire. Find the current I , as a function of k and R . You will probably need to do a two-dimensional integral in polar coordinates. (In two-dimensional polar coordinates, the area element is $dS = r dr d\theta$.)

3. A copper conductor has 8.5×10^{28} electrons per cubic meter available to carry current, i.e., the carrier density is $8.5 \times 10^{28} \text{m}^{-3}$. The drift velocity is found to be $3 \times 10^{-6} \text{m/s}$.

- (a) [2 pts] Look up and report the charge of an electron in SI units.
(b) [4 pts] Calculate the current density J .

4. [6 pts] The current density in some region is found to be

$$\mathbf{J} = \frac{y}{\sqrt{x^2 + y^2}} \hat{i} - \frac{x}{\sqrt{x^2 + y^2}} \hat{j}$$

Use the continuity equation to show that the charge density in this region does not change with time.

5. A zinc wire sample at 20°C is found to have resistivity $\rho = 5.9 \times 10^{-8} \text{ohm-m}$.

- (a) [3 pts] Find the conductivity σ .
(b) [3 pts] An applied electric voltage drives a steady current through the zinc sample. The current density at a point in the sample is found to be $J = 4 \times 10^5 \text{A/m}^2$. Calculate the electric field at that point.
(c) [3 pts] The wire sample has cross-section area $4 \text{mm}^2 = 4 \times 10^{-6} \text{m}^2$ and length $1 \text{cm} = 10^{-2} \text{m}$. Find the resistance of the sample.